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 Studierichting: Informatica
 Oefeningenreeks 2E nr. 26.16

$$\begin{aligned}
 2 \cdot (\log_2(x-1) - \log_4(x^2+1)) &= 1 - \log_4 25 \\
 \Leftrightarrow 2 \cdot \left(\log_2(x-1) - \frac{\log_2(x^2+1)}{\log_2 4} \right) &= 1 - \frac{\log_2 25}{\log_2 4} \\
 \Leftrightarrow 2 \cdot \left(\log_2(x-1) - \frac{\log_2(x^2+1)}{2} \right) &= 1 - \frac{\log_2 25}{2} \\
 \Leftrightarrow 2 \cdot \log_2(x-1) - \log_2(x^2+1) &= 1 - \frac{\log_2 25}{2} \\
 \Leftrightarrow \log_2(x-1)^2 - \log_2(x^2+1) &= 1 - \frac{\log_2 25}{2} \\
 \Leftrightarrow \log_2 \left(\frac{(x-1)^2}{x^2+1} \right) &= 1 - \frac{\log_2 25}{2} \\
 \Leftrightarrow \log_2 \left(\frac{(x-1)^2}{x^2+1} \right) &= \log_2 2 - \frac{2 \log_2 5}{2} \\
 \Leftrightarrow \log_2 \left(\frac{(x-1)^2}{x^2+1} \right) &= \log_2 2 - \log_2 5 \\
 \Leftrightarrow \log_2 \left(\frac{(x-1)^2}{x^2+1} \right) &= \log_2 \frac{2}{5} \\
 \Leftrightarrow \frac{(x-1)^2}{x^2+1} &= \frac{2}{5} \\
 \Leftrightarrow (x-1)^2 &= \frac{2}{5}(x^2+1) \\
 \Leftrightarrow x^2 - 2x + 1 &= \frac{2}{5}x^2 + \frac{2}{5} \\
 \Leftrightarrow 5x^2 - 10x + 5 &= 2x^2 + 2 \\
 \Leftrightarrow 3x^2 - 10x + 3 &= 0 \\
 D &= b^2 - 4ac \\
 &= 100 - 4 \cdot 3 \cdot 3 \\
 &= 100 - 36 \\
 &= 64 \\
 x_1 &= \frac{-b + \sqrt{D}}{2a} = \frac{10 + 8}{6} = \frac{18}{6} = 3 \\
 x_2 &= \frac{-b - \sqrt{D}}{2a} = \frac{10 - 8}{6} = \frac{2}{6} = \frac{1}{3}
 \end{aligned}$$

$$V = \{3\}$$

References